



OPEN Enhancing indoor monitoring of visually impaired people using temporal convolutional network with optimization model in IoT environment

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The Internet of Things (IoT) has emerged as a powerful technology in various fields, including healthcare, assisting the elderly and disabled individuals. Solution-based IoT is widely utilized in healthcare support in diverse aspects of their daily lives and activities. These IoT-based assistive systems are crucial in uplifting the quality of life for disabled individuals and older adults. Furthermore, IoT-based interior navigation methods assist with easy accessibility and movement in areas, ensuring that disabled people can find objects with greater ease. Machine learning (ML) and deep learning (DL) approaches are efficiently beneficial for detecting indoor objects for visually challenged people. This paper proposes the Metaheuristic Optimization-Driven Attention Mechanism for Enhancing Indoor Monitoring of Visually Impaired People (MOAM-EIMVIP) technique. The main intention of the MOAM-EIMVIP technique is to develop an IoT-based DL model for indoor monitoring of disabled individuals. Initially, the data pre-processing stage applies min-max normalization to convert input data into a suitable format. The sine-cosine algorithm (SCA) is employed to select the feature process to detect and choose the most relevant features from input data. Moreover, the temporal convolutional network-attention mechanism (TCNA) model is implemented for classification. Finally, the Eurasian oystercatcher optimizer (EOO) method alters the hyperparameter tuning of the TCNA model optimally and results in higher classification execution. An extensive simulation is performed to highlight the performance of the MOAM-EIMVIP approach under the HAR dataset. The experimental analysis of the MOAM-EIMVIP approach portrayed a superior accuracy value of 99.30% over existing methods.

Keywords Attention mechanism, Eurasian oystercatcher optimizer, Visually impaired people, IoT, Indoor monitoring, Feature selection

IoT has a substantial capability to improve the existence of disabled individuals. By combining technologies and smart gadgets, the IoT provides a novel solution to tackle several challenges experienced by disabled individuals and promote inclusiveness¹. Using IoT-enabled assistive applications and devices, people with disability can attain amplified independence, admittance to crucial services, and enhanced living standards. IoT gadgets with actuators, connectivity and sensors allow real-world assistance and monitoring in medical care, mobility and accessibility². For example, wearable gadgets can track essential signs and transfer data to medical care experts, offering early clinical support and personalized care³. Smart home technologies can automate jobs, creating living space available and allowing people to regulate their settings over voice commands or another adaptable interface. Likewise, IoT-based navigation methods promote simple movements and availability in public spaces, ensuring that disabled individuals can navigate buildings, streets and transportation⁴. Whereas IoT provides immense possibilities, it is vital to tackle security and confidentiality issues to safeguard the sensitive data gathered by these gadgets⁵.

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Although a global positioning system (GPS) provides a sufficient indoor and outdoor localization solution, it is still an open challenge. Various hardware and algorithmic methods are based on the preferred cost and accuracy. The accessible localization method offers position-related data, guidance and advertisements in significant indoor places⁶. These methods are also employed for asset tracking in industrial settings. Some critical localization applications are familiar to clinical purposes. Nevertheless, the latest developments in IoT applications have proven they are expensive for medical care monitoring⁷. A novel perception of indoor localization methods is generated, which is the co-relation of indoor localization information with the health care status of people, particularly for disabled people. The indoor positioning system (IPS) introduces real-time localization of objects or people, and spaces are enveloped by different settings, employing a network for transmitters and receivers⁸. DL and ML have made meaningful challenge feature extractors from the raw sensor data⁹. Persons with cognitive or physical impairments frequently face challenges handling day-to-day jobs, including controlling their home setting. With the growth of smart technologies and connected devices, there is an increasing opportunity to progress systems which provide autonomy and assistance to people with disability¹⁰.

This paper proposes the Metaheuristic Optimization-Driven Attention Mechanism for Enhancing Indoor Monitoring of Visually Impaired People (MOAM-EIMVIP) technique. The main intention of the MOAM-EIMVIP technique is to develop an IoT-based DL model for indoor monitoring of disabled individuals. Initially, the data pre-processing stage applies min-max normalization to convert input data into a suitable format. The sine-cosine algorithm (SCA) is employed to select the feature process to detect and choose the most relevant features from input data. Moreover, the temporal convolutional network-attention mechanism (TCNA) model is implemented for classification. Finally, the Eurasian oystercatcher optimizer (EOO) method alters the hyperparameter tuning of the TCNA model optimally and results in higher classification execution. An extensive simulation is performed to highlight the performance of the MOAM-EIMVIP approach under the HAR dataset. The key contribution of the MOAM-EIMVIP approach is listed below.

- The MOAM-EIMVIP model utilizes min-max normalization to scale the input data to a consistent range, enhancing the learning process's stability and convergence. This confirms that all features contribute equally, preventing dominant features from altering the outputs. It improves overall performance, specifically in models sensitive to varying data distributions.
- The MOAM-EIMVIP method employs the SCA technique to choose the most relevant features from high-dimensional data, confirming that only the most informative attributes are retained. This mitigates the dimensionality, enhancing model efficiency and computational speed. By removing irrelevant features, the approach also assists in averting overfitting and improving generalization.
- The MOAM-EIMVIP approach integrates TCN with an attention mechanism for classification, effectively capturing sequential dependencies in the data. The attention mechanism highlights the most critical time steps, improving the model on key information. This integrated approach enhances the accuracy of the classification by improving both temporal learning and context-aware decision-making.
- The MOAM-EIMVIP methodology implements the EOO model for hyperparameter tuning, replicating the foraging behaviour of the EO technique to explore the search space effectively. This method ensures a robust and effective optimization process, resulting in the selection of optimal hyperparameters. As a result, the model's performance is significantly improved through fine-tuned settings.
- The MOAM-EIMVIP model's novelty is its hybrid integration of bio-inspired optimization with DL, combining SCA for feature selection and EOO for hyperparameter tuning with TCNA-based classification. This unified architecture utilizes intelligent search strategies with powerful sequence modelling to improve accuracy and adaptability. Its innovative design enables effective handling of high-dimensional, time-dependent data. Such synergy between optimization and learning techniques makes the model uniquely robust and performance-driven.

Related works

Nguyen et al.¹¹ present an innovative, low-cost smart wheelchair method. The presented model employs a reasonable webcam and an Nvidia jetson nano-embedded computer to recognize and process six diverse hand gestures to aid wheelchair navigation. This method leverages the YOLO-v8n technique. Feature extract ROI of the hand gesture and background, considerably enhancing the precision of gesture recognition. Lee and Zhang¹² progress an innovative artificial intelligence of things (AIoT)-based structure for forecasting multi-dimensional IEQ. The presented structure combines IoT with DL models to systematically integrate multi-dimensional IEQ data and multi-modal feedback from occupants. Alfahad et al.¹³ advance an auto-adjusting model to choose coefficients dependent on the modern automatic RSSI class. This model is accepted by the IoT incorporated with BLE to accomplish these objectives. The advanced method comprises data aggregation, storage, Bluetooth beacons and a web interface for real-world visualization and tracking. The RSSI footprinting technique is accepted to identify a specific zone with IEQ. Alzahrani¹⁴ projected an innovative TCN-based fall activity recognition system for disabled persons (TCN-FARSDP) methodology. This image pre-processing is accomplished by employing GF to remove noise and enhance the clarity of the acquired image. Afterwards, the ensemble of feature extractor techniques contains three methodologies: DenseNet121, MobileNetV3Large, and NASNetMobile. A TCN classifier is employed to recognize fall activities. Finally, fine-tuning is executed using the Adamax to improve the model's stability and convergence. Li et al.¹⁵ projected an intellectual communication method for BIM techniques that depends on gesture recognition. To aid intuitive and natural human-machine communication among BIM and gesture techniques, it has custom-designed six targeted gestures for HBIM. Moreover, a CNN-based gesture recognition model was advanced to recognize gesture features. Eventually, several operations are applied to the BIM technique. Qin et al.¹⁶ developed a DL-based solution for identifying

abnormal neck behaviour. They suggested a method named NABNet, which integrates OD reliant on YOLO-v5s with pose estimation depending on Lightweight OpenPose. This technique employed NABNet on the cloud and edge gadgets.

Wu et al.¹⁷ present TBDNet, combining tracking-before-detect (TBD) from traditional signal processing into learning-based techniques. TBDNet emulates first-track-then-detect pipeline. It maps input heatmap sequences to higher-level frame-to-frame aspects to adjust targets' time-varying intensity distribution and motion pattern. Sampath et al.¹⁸ precisely estimate indoor positions by integrating radio frequency identification (RFID) data with a deep q-learning (DQL) and adaptive emperor penguin colony optimizer (AEPCO) model, improved by Gaussian filtering (GF) technique to enhance the quality of training and eliminate received signal strength indicator (RSSI) noise. Thandu and Rathinam¹⁹ improve indoor navigation for VIPs by utilizing an AI chatbot powered by natural language processing (NLP) and volcano eruption optimization (VEO) for real-time, accessible guidance. Dikkatwar et al.²⁰ analyze and rank wearable assistive technologies for visually impaired people (VIP), emphasizing capabilities, user-informed metrics, and future research directions. Valipour, de Antonio, and Cabrera²¹ propose a user-centred analysis and design framework for indoor scene understanding assistive solutions for VIPs, addressing key challenges and giving guidelines for choosing appropriate technologies. Afif et al.²² predict optimal indoor temperature for occupant comfort by incorporating IoT-based monitoring with multivariate statistical analysis and locally weighted regression. Saraswathi et al.²³ developed a voice-controlled app utilizing the DL model to analyze and describe scenes, with real-time on-device processing, feature extraction, and audible feedback for visually impaired users. Paul et al.²⁴ developed a smart assistive stick integrating ultrasonic sensors, GPS, global system for mobile communications (GSM), and Artificial Intelligence (AI)-based object detection to improve independent mobility and safety for the VIP. Pydala, Kumar, and Baseer²⁵ introduce VISISENSE. This IoT-based assistive technology system improves navigation support for the visually impaired through advanced object detection, edge computing, and cloud-based computer vision models. Hossain, Anjom, and Chowdhury²⁶ developed a lightweight U-Net-based methodology with obstacle distance detection for real-time footpath navigation, enabling visually impaired users to receive audio-guided directions via an Android application. Nandhini et al.²⁷ compare three object detection algorithms, namely YOLOv3 for image-based object detection, OpenCV for video-based detection without action recognition, and CNN for real-time action detection with improved accuracy.

Despite the improvements in assistive technologies for VIPs, various limitations remain. Many existing models depend on complex and expensive hardware, restricting their accessibility and affordability for users. Additionally, while some systems integrate AI and DL, they predominantly suffer from high computational requirements, which affect real-time performance, particularly on mobile or embedded devices. Furthermore, existing object detection and activity recognition methods may not always attain the desired accuracy under diverse environmental conditions, resulting in potential safety concerns. Moreover, existing systems often lack adaptability to diverse indoor environments and fail to give comprehensive, context-aware assistance. There is also a requirement for more effectual and scalable solutions that provide seamless integration across various devices and settings while ensuring user comfort and accuracy in navigation and interaction.

Materials and methods

This manuscript proposes the MOAM-EIMVIP technique. The primary intention of the method is to develop an IoT-based DL model for indoor monitoring of disabled individuals. Figure 1 defines four stages: data normalization, dimensionality reduction, activity recognition using TCNA, and parameter selection using EEO.

Stage I: min-max normalization

At first, the data pre-processing stage applies min-max normalization to convert input data into a proper format²⁸. This normalization is chosen as it scales the overall features to a fixed range, usually between 0 and 1, ensuring uniformity across variables with diverse units or scales. This normalization model is effectual when the data is not normally distributed, as it does not make assumptions about the underlying distribution. Rescaling the data enhances ML models' convergence speed and stability, particularly gradient-based methods. Moreover, min-max normalization is computationally efficient and easy to implement, making it an ideal choice for pre-processing massive datasets in DL and other ML applications.

Data normalization is an essential stage in data pre-processing. It involves converting data into a standard form to decrease data complexity, increase data quality, and remove redundancy. This method typically standardizes numeric data by scaling it to a constant set of values, making it simpler to analyze and compare.

Min-max normalization, or feature scaling, is utilized to change the initial data linearly. This model attained each scaled data point among (0,1). This might be completed using the succeeding Eq. (1): Over min-max normalization, the links among the values of the new data are preserved.

$$Z^* = \frac{z - \min(z)}{\text{range}(z)} = \frac{z - \min(z)}{\max(z) - \min(z)} \quad (1)$$

Meanwhile, $\text{range}(z)$ signifies the range between minimum and maximum, and $\min(z)$ represents the minimum value. The Z^* range falls in the interval of [0,1], and the width of the interval is 1.

Stage II: SCA-based feature selection

For the selection of feature process, the SCA technique is utilized to detect and choose the most relevant features from input data²⁹. This technique is preferred because it can effectively explore the search space and select the most relevant features, enhancing the model's accuracy and performance. The characteristics of the SCA model

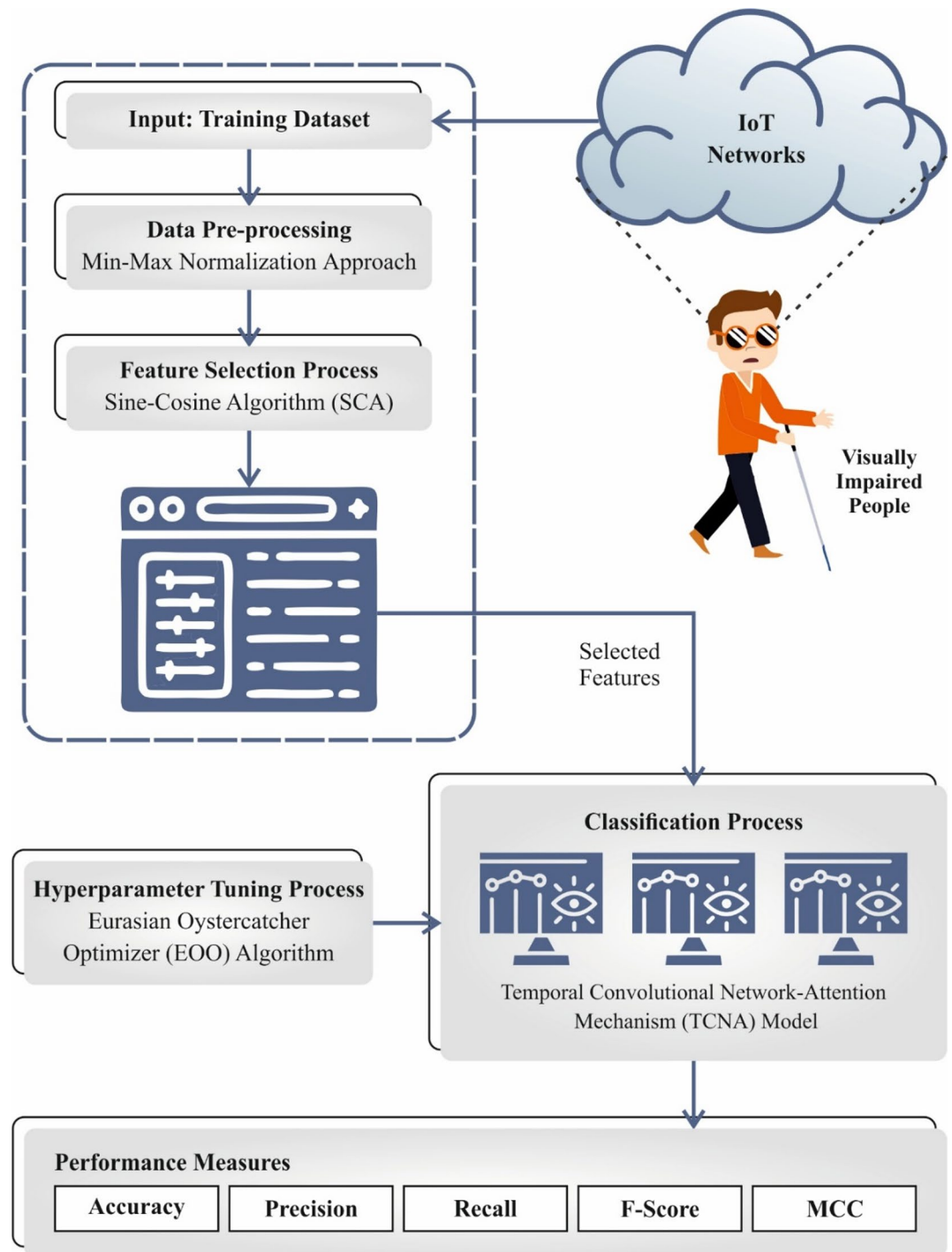


Fig. 1. Overall process of MOAM-EIMVIP model.

replicate the natural sine and cosine functions to balance exploration and exploitation during optimization, making it robust for high-dimensional and complex datasets. Unlike conventional feature selection methods, the SCA model does not depend on gradient information, making the technique appropriate for non-linear issues. Its global search capability also confirms that it can avoid local optima, enhancing generalization. Additionally, SCA's simplicity and ease of implementation make it an ideal choice over other optimization techniques. Figure 2 illustrates the steps involved in the SCA methodology.

SCA is an original population-based heuristic model that begins by creating numerous random solutions and then assists them in diverging or converging with the best performance. In addition, to highlight the exploitation and exploration of the searching region, this model combines a combination of adaptive and random variables. The stochastic population-based optimizer usually contains dual main stages, exploitation and exploration, and regarding SCA, each of these stages is combined into the following location-updated equations. At the same

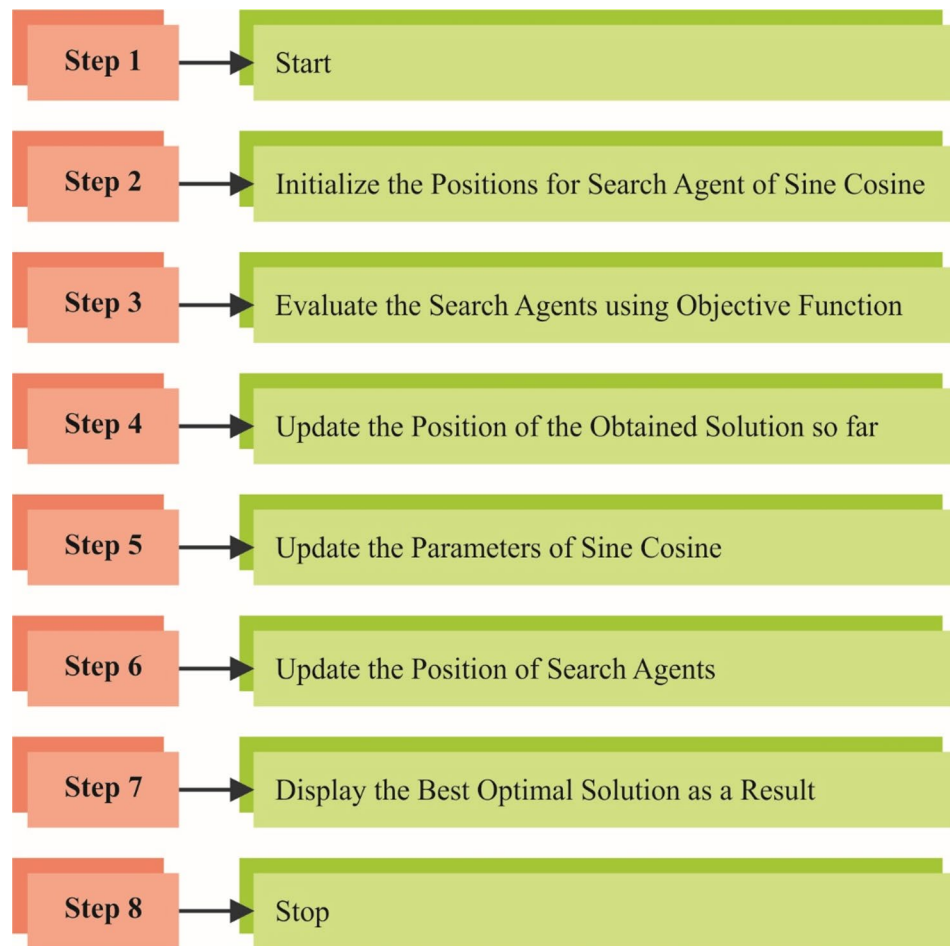


Fig. 2. Steps involved in the SCA approach.

time, X_j^t and P_j^t denote the location of the present solution and destination point in the size at t^{th} iteration correspondingly. r_1 , r_2 and r_3 denote random numbers and specify complete values as in Eq. (2).

$$X_j^{t+1} = \begin{cases} X_j^t + r_1 \cdot \sin(r_2) \cdot |r_3 \cdot P_j^t - X_j^t|, & \text{if } r_4 < 0.5 \\ X_j^t + r_1 \cdot \cos(r_2) \cdot |r_3 \cdot P_j^t - X_j^t|, & \text{if } r_4 \geq 0.5 \end{cases} \quad (2)$$

The range for the resulting location, signified by r_1 , is stated within the range $[2, 2]$ in Eq. (2) above. In this paper, the degree of movement away from or toward the destination was established by r_2 , which is between 0 and 2π . Moreover, r_3 adds the stochastic random weight to the end, which may improve ($r_3 > 1$) or reduce ($r_3 < 1$) the distance. Finally, r_4 is restricted to a range of zero and one, establishing an equivalent transition between sine and cosine functions or reverse.

In Eq. (2), a different area is determined among dual possible solutions inside the searching area. These solutions retain the ability to explore away from the limitations of their specific destination by fine-tuning the magnitudes of sine-cosine functions. This dynamic adaptation ensures that the large searching region is successfully discovered. Additionally, for sine-cosine functions to be regular, it is promising to constantly move one solution nearer to the other, enabling exploitation.

The following equation is applied to dynamically adjust the sine and cosine function sizes in Eq. (3) to attain the model's balanced exploitation and exploration stage.

$$r_1 = a - \frac{t \cdot a}{T} \quad (3)$$

In such a case, a represents fixed constants, t for the present iteration amount, and T for the maximal iteration counts.

The fitness function (FF) deliberates the classification precision and the chosen feature counts. It maximizes the classification precision and lowers the set size of the preferred attributes. Consequently, the succeeding FF is implemented to evaluate specific solutions, as represented in Eq. (4).

$$Fitness = \alpha \bullet ErrorRate + (1 - \alpha) \bullet \frac{\#SF}{\#All_F} \quad (4)$$

Here, *ErrorRate* denotes the classification error ratio that employs the selected features. *ErrorRate* is measured as the incorrect percentage categorized to the sum of classifications completed, stated as a value among (0,1), *#SF* signifies designated feature counts, and *#All_F* refers to the total number of features in the novel dataset. α is applied for controlling the significance of subset length and classification quality.

Stage III: activities recognition using the TCNA model

Moreover, the proposed MOAM-EIMVIP technique applies the TCNA model for classification. This model is chosen due to its ability to effectively capture sequential dependencies in time-series data, which is significant for walking, sitting, or standing. The attention mechanism confirms that crucial time steps attain higher weight, while the convolutional layers allow the model to learn hierarchical temporal features, improving accuracy. Unlike conventional RNNs, this model handles more extended sequences without the vanishing gradient problem, resulting in improved recognition of complex activities. Furthermore, TCNA is computationally effectual and capable of scaling to large datasets, making it an ideal choice for real-time activity recognition tasks. Its capability to concentrate on key patterns in time-series data sets it apart from other models that treat all time steps equally. Figure 3 depicts the architecture of the TCNA model.

TCN is a standard NN method that combines causal and dilated convolutions and the residual component³⁰. It is applied for prediction tasks by modelling sequential data, permitting the capture of temporal dependences using their dilated causal convolutions and wide-ranging receptive areas. Equation (5) determined the convolutional process during this model.

$$F(p) = (xf_{cl})(p) = \sum_{i=0}^{k-1} f(i) \bullet x_{(p-di)} \quad (5)$$

Meanwhile, k, d, p, di characterize the convolutional kernel size, expansion coefficient, and consistent sequence within the convolutional kernel. TCN precision by particularly concentrating on the most related features or time steps inside the sequence of input, the TCNA technique was used. This permits the method to dynamically weigh significant information, enhancing its capability to capture longer-term dependences and composite temporal patterns. By allocating weighted values to all variables, the attention mechanism concentrated on significant variables to increase the forecast outcome. Eqs identify the complete evaluation of the output attention vectors. (6) and (7).

$$G_i^d = \frac{\exp(g_i)}{\sum_{k=1}^m \exp(g_k)} \quad (6)$$

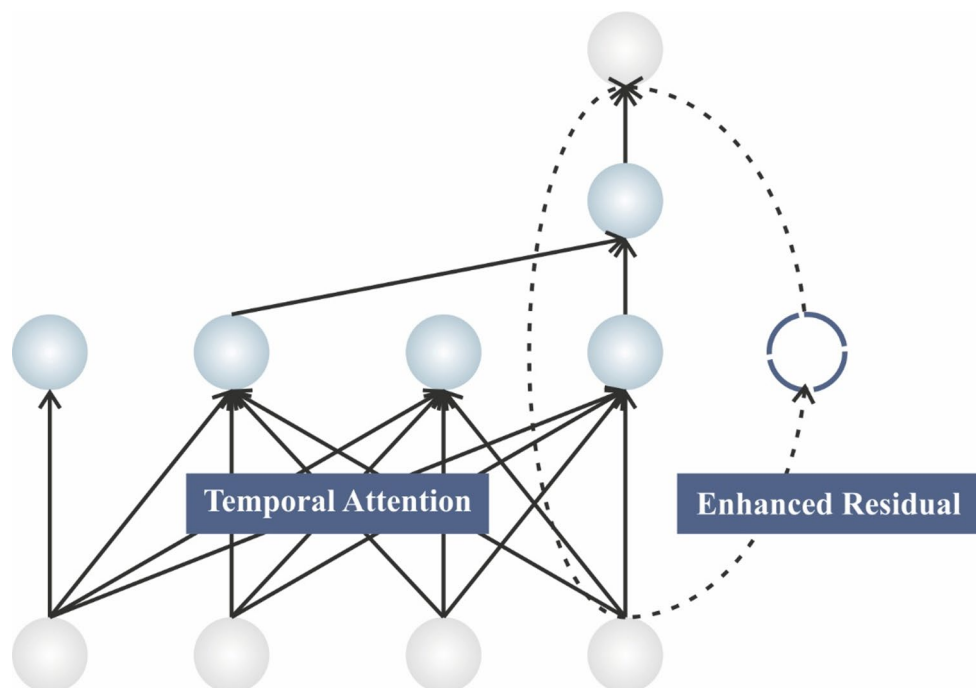


Fig. 3. Architecture of TCNA model.

$$\beta_j = \frac{\exp(g_j^d)}{\sum_{k=1}^n \exp(g_k^d)} \quad (7)$$

whereas the $[gl, \dots, gm]$, β_j , n denotes the block vector of the resultant vector, the weight coefficient, and the output vector length.

Stage IV: parameter selection using EOO

Finally, the EOO approach optimally alters the hyperparameter tuning of the TCNA model and results in higher classification performance³¹. This model was chosen due to its robust global search capability, inspired by oystercatcher birds' foraging behaviour. This model efficiently balances exploration and exploitation, enabling it to navigate intrinsic, high-dimensional parameter spaces without getting trapped in local optima. Compared to conventional techniques such as grid or random search, EOO presents faster convergence and improved optimization of hyperparameters. The model's flexibility also plays a crucial part in making the model highly appropriate for both continuous and discrete parameter tuning. Furthermore, the nature-inspired strategy of the model allows it to adapt to diverse problem landscapes, improving model accuracy and generalization. This makes it an ideal and effectual choice for fine-tuning DL models. Figure 4 represents the workflow of the EOO method.

The food behaviours of Eurasian oystercatchers (EO) while searching for mussel's act as the stimulus to EOO. It was created by detecting how the Oystercatchers eat and how they perform. Discover the correct solutions relies on discovering the proper balance among calories consumed, energy produced, and time loss. This

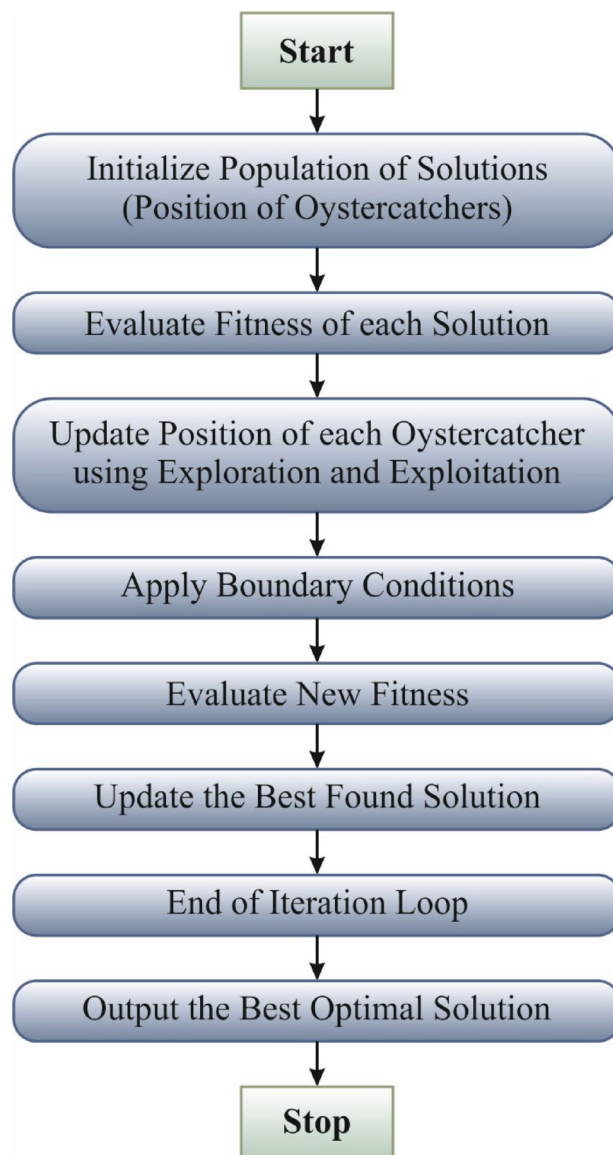


Fig. 4. Working flow of the EOO approach.

searching approach and the related mussel selected by EO are presented as demonstrated. EO's main objective is to equalize energy and calories from the mussel. The mussel dimension, the energy spent on opening the mussel, and the calories gained from the mussel consumption are strongly related. The calorie count eaten and the time count required for opening mussels either improve by the rise in the mussel's size. Therefore, EO wastes essential energy in the initial process of separating mussels from their shells. Eqs exemplify the behaviour of EO in the exploration stage (Eqs. (8) and (9)).

$$A = B + D + G \bullet a \bullet (M_b - M_{h-1}) \quad (8)$$

The term A in Eq. (8) is the EO's energy, particularly even accessible lastly that is inverse with the iteration amount, G equates to the mussel size, and denotes randomly generated number amongst^{3,5} that specifies the border of the best size of mussel, and B represents the sum of time required for breaking the current mussel. Its value relies on G , D characterizes the energy needed currently, and a signifies a randomly generated number chosen to improve unpredictability and discover extra places within the searching space in the interval among $[0,1]$.

$$M_h = M_{h-1} \bullet I \quad (9)$$

Here, M_h signifies the position of the possible mussel. I designate the value of calories that the EO obtain according to the mussel dimension G . The period B is established according to Eq. (10).

$$B = \left(\frac{G - 3}{5 - 3} \right) \bullet 10 - 5 \quad (10)$$

Equation (10) uses the best mussel size G to calculate the time needed for breaking the mussel. Equation (11) establishes the present energy value accessible.

$$D = \left(\frac{h - 1}{c} \right) - 0.5; \text{ for } h > 1 \quad (11)$$

Equation (9) establishes the caloric quantity I , which depends on the mussel size G .

$$I = \left(\frac{G - 3}{5 - 3} \right) \bullet 2 + 0.6 \quad (12)$$

Here, the value of I is in the interval $[0.6, 0.8]$, while the outcomes from Eq. (10) provide outputs in the interval $[5, -5]$. These parameters are chosen for comprehensive testing. It is also crucial to note that when the time is negative, the time needed for opening the mussel outperforms the bird's capability, which is considered a limitation. In Eq. (11), the value D is obtained and decreases linearly from $[0.5, -0.5]$, where h represents the iteration value starting from H and ending at 1. The D value remains continuous during the last two iterations. Therefore, the terms B and D , illustrating the time and energy required to break the mussel, can have negative values. The B value in Eq. (8) and the I value in Eq. (9) rely on the randomized variable G , which changes continuously. This condition makes it feasible for EO to discover someplace within the solution area and prevent the best local problem, promising exploration every time. The main features of EOO, which assist in resolving optimizer difficulties, are conceptually described by the succeeding objectives:

- (1) The precision of selecting the mussels is calculated by utilizing the time needed for breaking a mussel, the mussel's size, and the bird's energy as features to evaluate the projected place of the preferred food.
- (2) In every cycle, the randomly generated numbers arrived in optimization to help explore novel places. Therefore, prevent a local minimal problem.
- (3) The random numbers used at each stage of the optimizer guarantee utilization and investigation.

Algorithm 1 demonstrates the EOO technique.

Step 1: Initialization

- Initialize the populace of oystercatchers (solutions). Each oystercatcher is represented as a vector of decision variables (solution).
- Define parameters for the optimization process:
 - Maximum iterations ($Tmax$)
 - Population size (N)
 - Search space boundaries (upper and lower boundaries for each decision variable)
 - Learning rate and other model-specific constants (e.g., $r1$, $r2$, α , etc.)

Step 2: Objective Function

- Define the objective function that the oystercatchers try to optimize (minimization or maximization). This objective function is utilized to compute the fitness of every solution.

Step 3: Foraging Behavior Update

- Each oystercatcher's position is updated depending on its prior position and the optimum position found by itself and other oystercatchers.

The general update rule can be described as follows:

$$X_i^{t+1} = X_i^t + r_1 \cdot \sin(r_2) \cdot |r_3 \cdot P_i^t - X_i^t|,$$

Where:

- X_i^{t+1} is the updated position of the i^{th} oystercatcher at time $t + 1$.
- X_i^t is the current position of the i^{th} oystercatcher at time t .
- $P_{the\ pi}^t$ is the position of the best solution found by the oystercatcher up to time t .
- r_1 , r_2 and r_3 are arbitrary numbers in $[0,1]$ that control the search direction and intensity.
- The term $|r_3 \cdot P_i^t - X_i^t|$ depicts the distance to the best solution found by the oystercatcher.

Step 4: Update Fitness

- Compute the fitness of the new position X_i^{t+1} utilizing the objective function.
- Update the best position P_i^t if the new position results in a better fitness value.

Step 5: Global Best Update

- Track the global best solution found so far across all oystercatchers.
- The global best position G^t is updated based on the best fitness value among all oystercatchers.

Step 6: Termination Condition

- Repeat steps 3 to 5 for a pre-defined number of iterations ($Tmax$) or until convergence is attained (based on a desired accuracy or stopping criteria).

Step 7: Output

- The algorithm outputs the global best solution found during the iterations as the final optimized solution.

Algorithm 1 EEO model.

Fitness selection is an essential factor in manipulating the execution of the EEO model. The hyper-parameter choice method contains the solution encoding model to calculate the efficiency of the candidate solutions. In this manuscript, the EEO model imitates accuracy as the main factor for designing the FF, which is exemplified as established.

$$Fitness = \max (P) \quad (13)$$

$$P = \frac{TP}{TP + FP} \tag{14}$$

Meanwhile, TP and FP illustrate the true and false positive values.

Performance validation

The performance analysis of the MOAM-EIMVIP approach is examined under the HAR dataset³². The technique is simulated using the Python 3.6.5 tool on PC i5-8600k, 250GB SSD, GeForce 1050Ti 4GB, 16GB RAM, and 1 TB HDD. The parameter settings are provided: learning rate: 0.01, activation: ReLU, epoch count: 50, dropout: 0.5, and batch size: 5. The dataset comprises 10,100 record samples with six classes defined in Table 1. The total no. of features is 561, and the selected features are 259.

Figure 5 presents the classifier outcomes of the MOAM-EIMVIP technique on the test database. Figure 5a and b exhibits the confusion matrices with truthful identification of 6 classes below 70%TRAPA and 30%TESPA. Figure 5c depicts the PR values, representing maximum performance across six classes. At last, Fig. 5d illustrates the ROC analysis, proving capable outcomes with high ROC outcomes for six classes.

Table 2; Fig. 6 display the indoor monitoring detection results of the MOAM-EIMVIP technique below 70%TRAPA and 30%TESPA. The results showed that the MOAM-EIMVIP technique was proficient in finding the samples. On 70%TRAPA, the MOAM-EIMVIP technique attains an average $accu_y$, $prec_n$, $reca_l$, F_{score} , MCC, and Kappa of 99.24%, 97.72%, 97.73%, 97.73%, 97.27%, and 97.26%. Followed by, on 30%TESPA, the MOAM-EIMVIP approach attains an average $accu_y$, $prec_n$, $reca_l$, F_{score} , MCC, and Kappa of 99.30%, 97.91%, 97.81%, 97.86%, 97.44%, 97.46%.

In Fig. 7, the TRA $accu_y$ (TRAAY) and validation $accu_y$ (VLAAY) results of the MOAM-EIMVIP models are demonstrated. The values of $accu_y$ are calculated across an interval of 0–30 epochs. The figure highlighted that the TRAAY and VLAAY values exhibit a growing trend, which indicates the proficiency of the MOAM-EIMVIP model with more excellent solutions through several iteration counts. Simultaneously, the TRAAY and VLAAY leftovers closer through the epochs, indicating lesser over-fitting and higher performance of the MOAM-EIMVIP model, securing consistent prediction on unknown instances.

Figure 8 illustrates the MOAM-EIMVIP approach’s TRA loss (TRALO) and VLA loss (VLALO). The loss values are computed over 0–30 epochs. The TRALO and VLALO values clarify the reduction of tendencies, which notifies the model’s capacity to balance a trade-off between data fitting and simplification. The constant decrease in loss values ensures the more significant outcomes of the MOAM-EIMVIP model and adjusts the prediction outcomes.

Table 3; Fig. 9 portray the overall comparison outcome of the MOAM-EIMVIP technique with other models^{33]–[34}. The results showed that the MOAM-EIMVIP technique achieved better performance. Based on $accu_y$, the MOAM-EIMVIP technique presents a higher $accu_y$ of 99.30%. At the same time, the Genetic Algorithm (GA), BiLSTM, Attention-LSTM, TAHAR-student-GRU, LeNet, ViT + CNN, and AlexNet models accomplish lower $accu_y$ of 91.92%, 96.04%, 95.51%, 89.65%, 96.70%, 98.06%, and 90.25%, correspondingly. At the same time, based on $prec_n$, the MOAM-EIMVIP technique provides a higher $prec_n$ of 97.91%, where the GA, BiLSTM, Attention-LSTM, TAHAR-student-GRU, LeNet, ViT + CNN, and AlexNet models achieve lesser $prec_n$ of 96.66%, 94.56%, 93.51%, 90.14%, 93.57%, 94.95%, and 95.01%, respectively. Moreover, concerning $reca_l$, the MOAM-EIMVIP methodology provides an improved $reca_l$ of 97.81%. In contrast, the GA, BiLSTM, Attention-LSTM, TAHAR-student-GRU, LeNet, ViT + CNN, and AlexNet approach attains lower $reca_l$ of 96.61%, 96.17%, 92.36%, 95.93%, 95.36%, 96.30%, and 92.35%, correspondingly. Additionally, based on the F_{score} , the MOAM-EIMVIP methodology provides a higher F_{score} of 97.86%. Likewise, the GA, BiLSTM, Attention-LSTM, TAHAR-student-GRU, LeNet, ViT + CNN, and AlexNet techniques attain minimal F_{score} of 93.02%, 96.80%, 91.70%, 95.94%, 90.33%, 90.84%, and 93.11%, correspondingly. Finally, based on MCC, the MOAM-EIMVIP methodology provides a higher MCC of 97.44%. Similarly, the GA, BiLSTM, Attention-LSTM, TAHAR-student-GRU, LeNet, ViT + CNN, and AlexNet techniques attains lesser MCC of 95.58%, 94.15%, 92.95%, 95.56%, 96.00%, 92.03%, and 95.32%, subsequently.

Table 4; Fig. 10 demonstrates the computational time (CT) analysis of the MOAM-EIMVIP technique with existing methods. Models such as GA and TAHAR-student-GRU exhibit higher CTs of 16.99 and 19.93 s respectively, showing potentially greater complexity or deeper architectures. On the contrary, models such

HAR database	
Classes	Records
“Walking”	1700
“Upstairs”	1500
“Downstairs”	1400
“Sitting”	1700
“Standing”	1900
“Laying Down”	1900
Total records	10,100

Table 1. Details of the dataset.

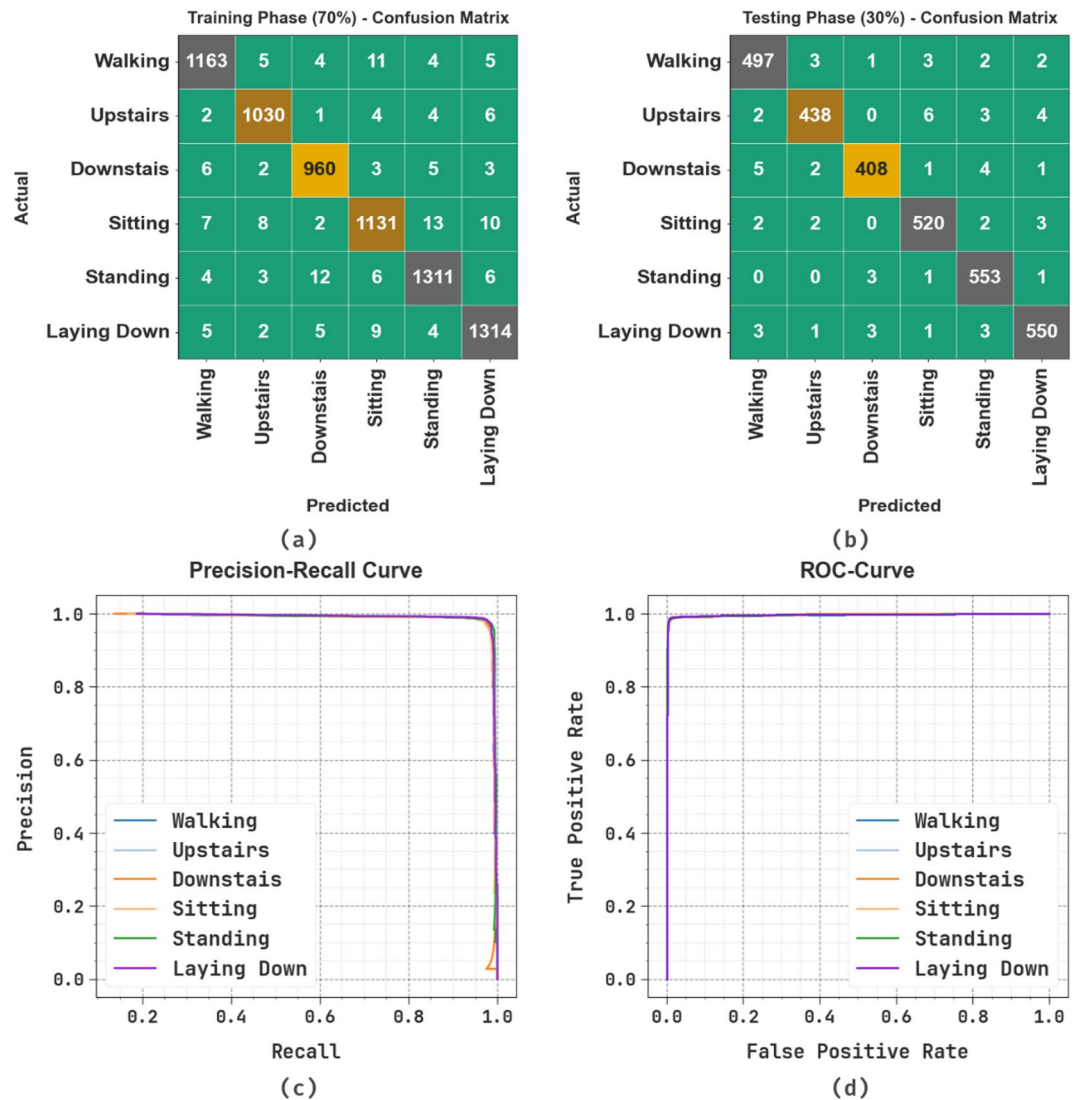


Fig. 5. (a,b) Confusion matrices and (c,d) curves of PR and ROC.

as BiLSTM and LeNet illustrate more moderate CT values of 11.7 and 13.52 s. The MOAM-EIMVIP model outperforms with the lowest CT of 6.74 s, illustrating its superior computational efficiency and making it highly appropriate for environments requiring faster CT. This analysis highlights the trade-off between model complexity and processing speed across diverse architectures.

Conclusion

In this paper, the MOAM-EIMVIP technique is proposed. The main intention of the MOAM-EIMVIP technique is to develop an IoT-based DL model for indoor monitoring of disabled individuals. Initially, the data pre-processing stage applies min-max normalization to convert input data into a suitable format. The SCA selects the most relevant features from input data to select the feature process. Moreover, the TCNA classifier is implemented for the HAR classification process. Finally, the EOO method optimally alters the TCNA model's hyperparameter tuning, resulting in higher classification execution. An extensive simulation is performed to highlight the performance of the MOAM-EIMVIP approach under the HAR dataset. The experimental analysis of the MOAM-EIMVIP approach portrayed a superior accuracy value of 99.30% over existing methods. The limitations of the MOAM-EIMVIP approach comprise restricted generalization across diverse real-world scenarios due to the constrained nature of the dataset. The scalability of the system under high data throughput or large-scale deployment remains unexplored. Real-time responsiveness and adaptability to growing input patterns were not evaluated. User personalization features are minimal, which may affect accessibility for a broader user base. Furthermore, the interpretability of the model outputs could be enhanced for better trust and transparency. Future works can explore integration with adaptive feedback systems, extend analysis on cross-domain datasets, improve user-centered design, and improve interpretability for practical deployment.

Class labels	$Accu_y$	$Prec_n$	$Recal_l$	F_{Score}	MCC	Kappa
TRAPA (70%)						
Walking	99.25	97.98	97.57	97.77	97.32	97.39
Upstairs	99.48	98.10	98.38	98.24	97.93	97.99
Downstairs	99.39	97.56	98.06	97.81	97.46	97.53
Sitting	98.97	97.16	96.58	96.87	96.26	96.31
Standing	99.14	97.76	97.69	97.73	97.19	97.09
Laying Down	99.22	97.77	98.13	97.95	97.47	97.26
Average	99.24	97.72	97.73	97.73	97.27	97.26
TESPA (30%)						
Walking	99.24	97.64	97.83	97.74	97.28	97.16
Upstairs	99.24	98.21	96.69	97.44	97.00	97.06
Downstairs	99.34	98.31	96.91	97.61	97.23	97.29
Sitting	99.31	97.74	98.30	98.02	97.60	97.68
Standing	99.37	97.53	99.10	98.31	97.93	97.91
Laying Down	99.27	98.04	98.04	98.04	97.59	97.64
Average	99.30	97.91	97.81	97.86	97.44	97.46

Table 2. Indoor monitoring detection of MOAM-EIMVIP technique under 70%TRAPA and 30%TESPA.

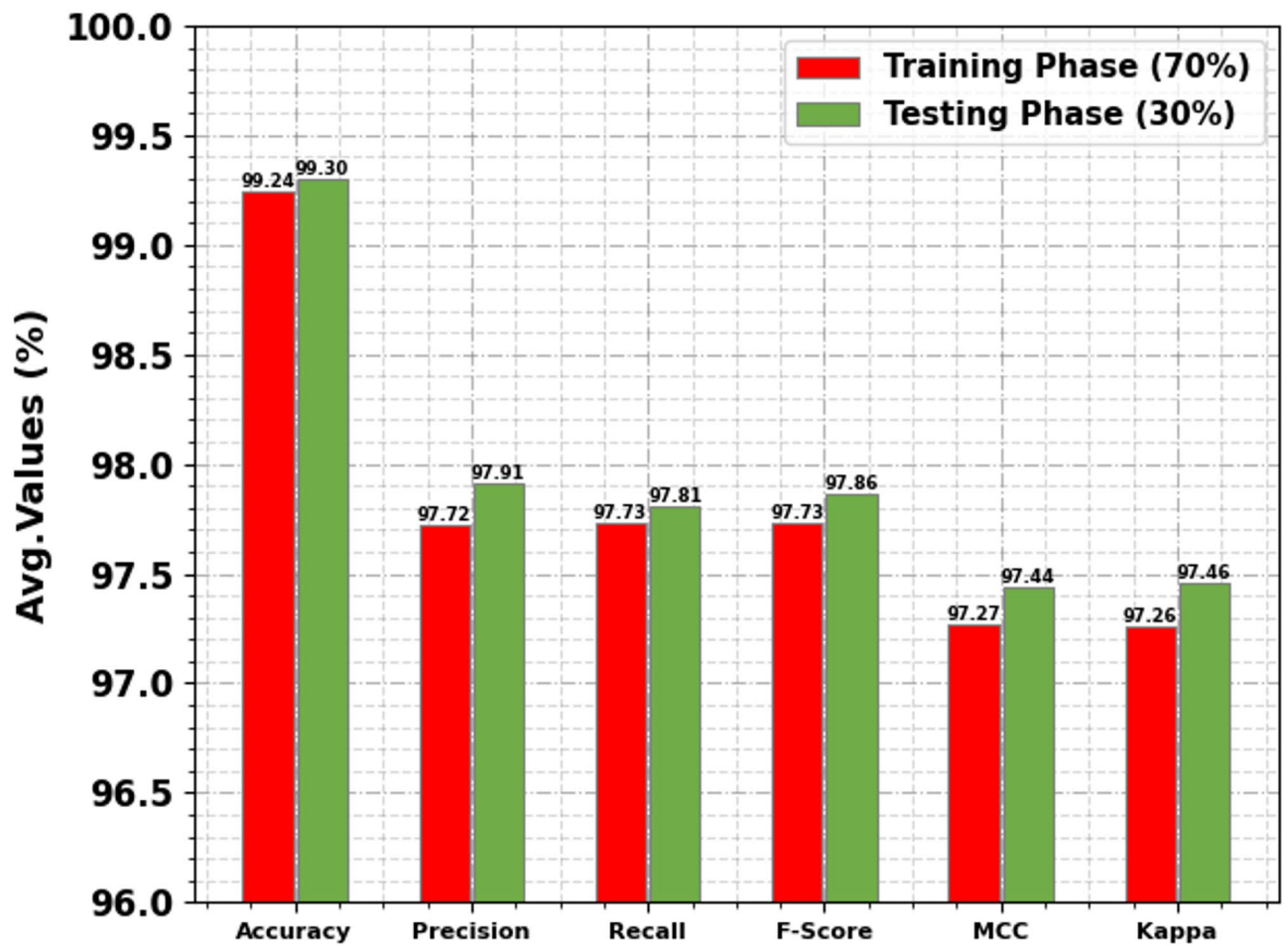


Fig. 6. Average of MOAM-EIMVIP technique below 70%TRAPA and 30%TESPA.

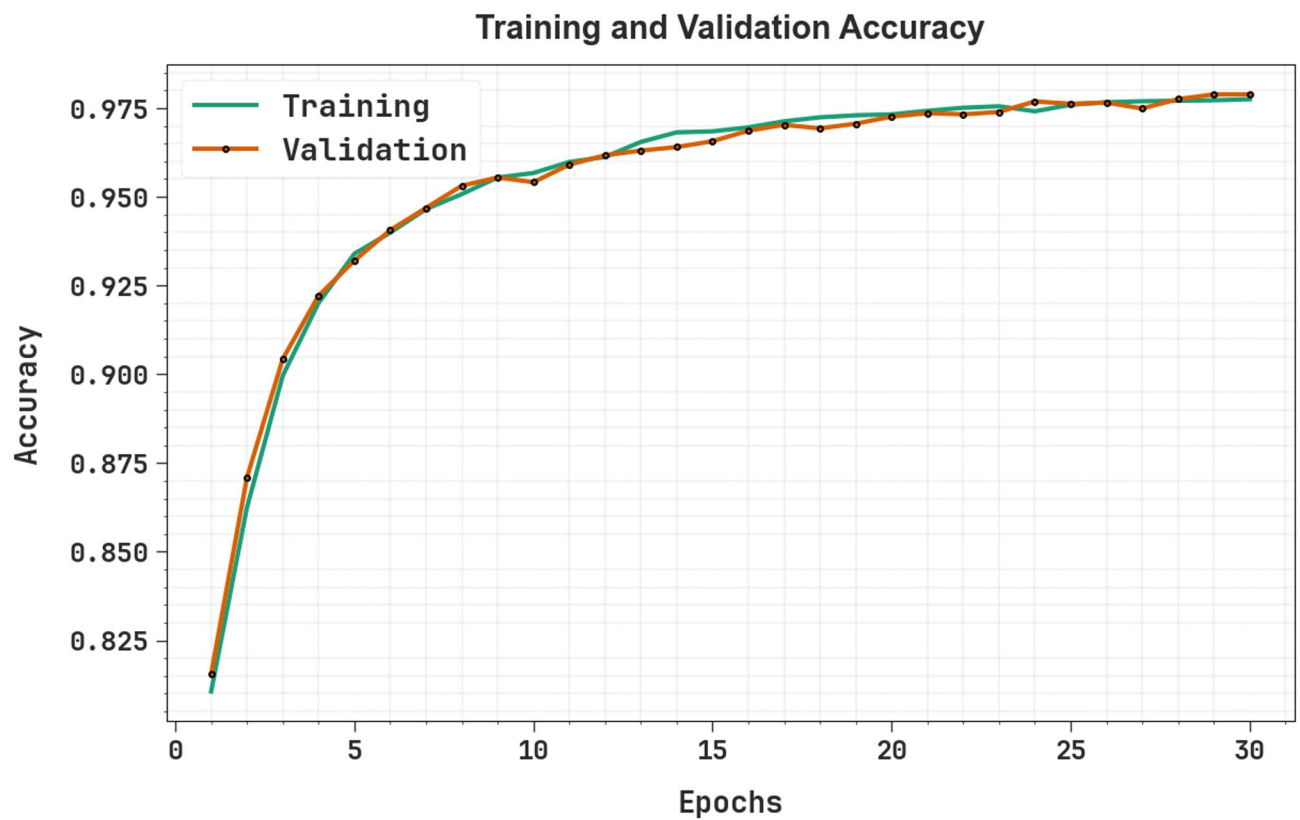


Fig. 7. $Accu_y$ analysis of the MOAM-EIMVIP technique.

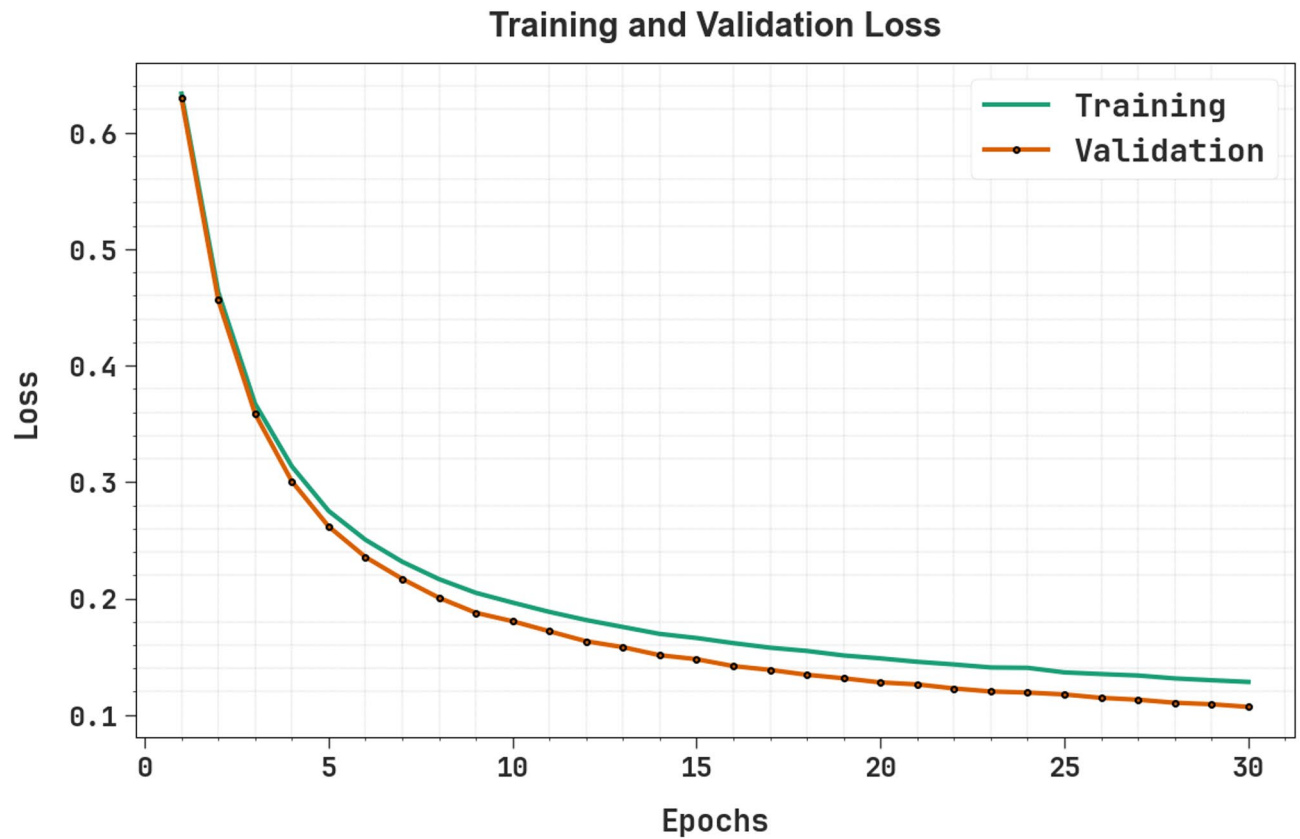


Fig. 8. Loss graph of the MOAM-EIMVIP technique.

Method	$Accu_y$	$Prec_n$	$Recal$	F_{Score}	MCC
GA	91.92	96.66	96.61	93.02	95.58
BiLSTM	96.04	94.56	96.17	96.80	94.15
Attention-LSTM	95.51	93.51	92.36	91.70	92.95
TAHAR-student-GRU	89.65	90.14	95.93	95.94	95.56
LeNet Model	96.70	93.57	95.36	90.33	96.00
ViT + CNN Method	98.06	94.95	96.30	90.84	92.03
AlexNet	90.25	95.01	92.35	93.11	95.32
MOAM-EIMVIP	99.30	97.91	97.81	97.86	97.44

Table 3. Comparison study of the MOAM-EIMVIP methodology with existing models³³–³⁴.

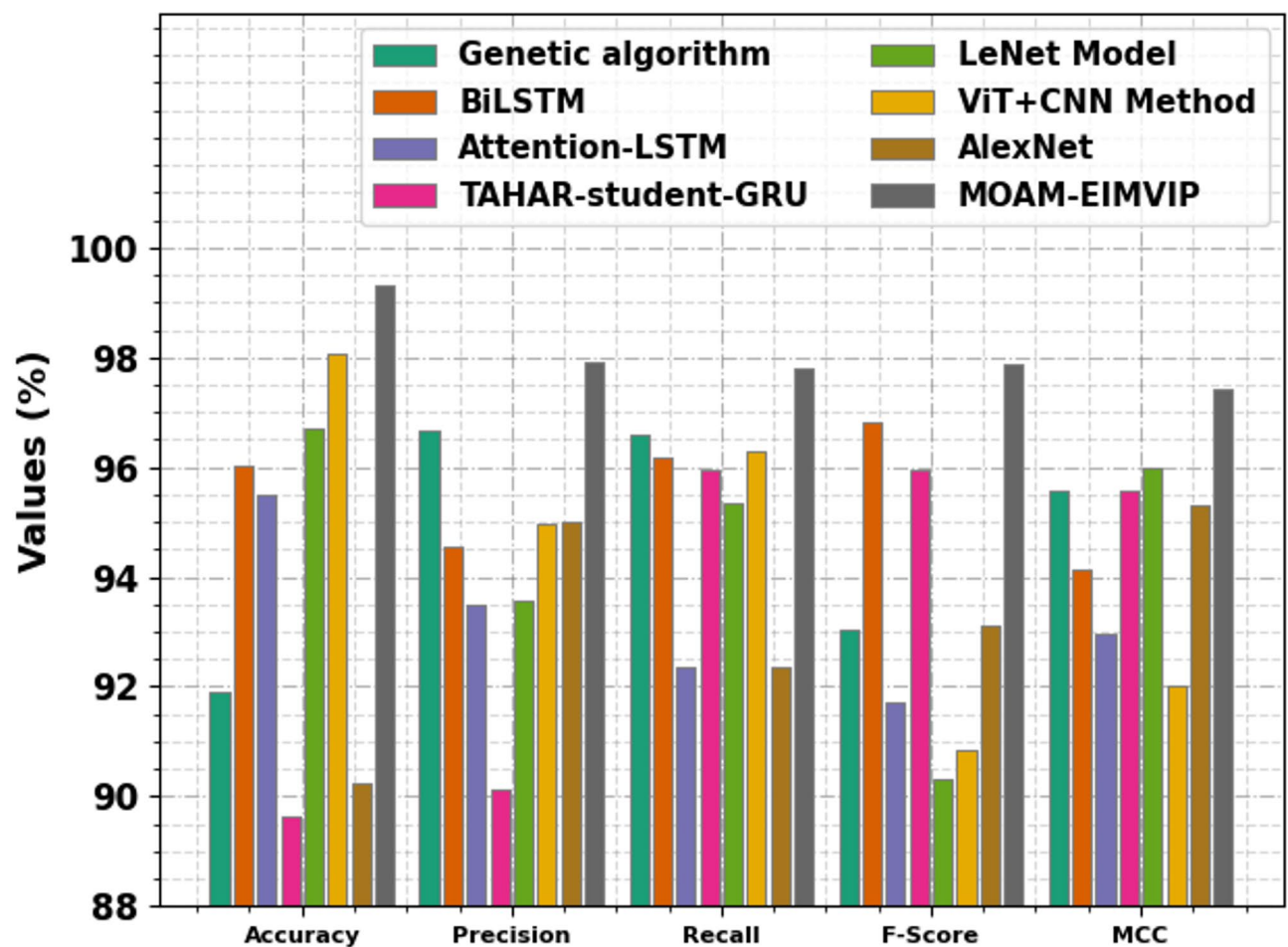


Fig. 9. Comparison study of the MOAM-EIMVIP methodology with existing models.

Method	CT (sec)
GA	16.99
BiLSTM	11.7
Attention-LSTM	16.82
TAHAR-student-GRU	19.93
LeNet Model	13.52
ViT + CNN Method	15.13
AlexNet	15.64
MOAM-EIMVIP	6.74

Table 4. CT evaluation of the MOAM-EIMVIP technique with existing methods.

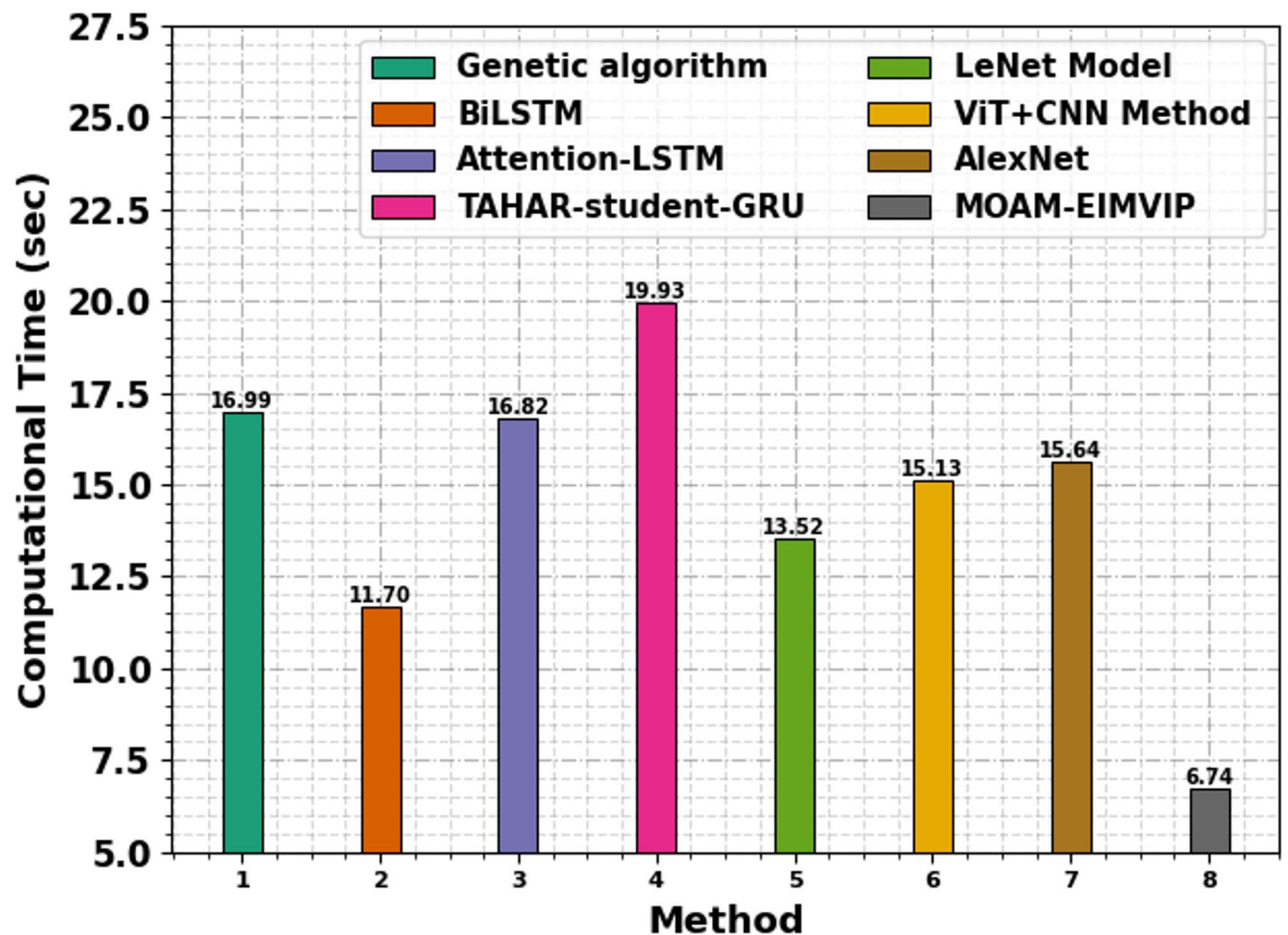


Fig. 10. CT evaluation of the MOAM-EIMVIP technique with existing methods.

Data availability

The data supporting this study's findings are openly available in the Kaggle repository at <https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones>, reference number³².

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Author contributions

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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